

Hybrid HDDs

Seagate has released hybrid HDDs with consisting of HDD platters for storage, 32MB DRAM cache and 4 GB single layer NAND flash cache



Evolution Of The Senses

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HDTVs have become something of a common sight, all you need to do is pop into a large electronics mall and you're sure to see some of these beauties on display. Larger panels are gaining popularity thanks to falling prices, as manufacturers adopt better technologies and reap the benefits of larger scale production. Despite this, some vendors continue to push smaller 19, 22, and 26-inch displays shamelessly, but the prices of larger panels have fallen to ridiculous levels. For example, some 42-inch HDTVs are now costing less than their 32-inch siblings did around 18 months back.

The larger a screen gets, the more immersive your experience gets, and

this is why people still flock to cinema theatres to watch movies – the super huge display. However, with 47 and 52-inch displays becoming available for well under a lakh, the place for cinema-like immersion may well be your living room. Add a third dimension to the mix, and things can only get better. Imagine watching compatible movies on a 3D screen – the action will seem to jump out at you. At the moment, 3D displays are a very new, and remember that while it's infinitely cool to own one at the moment, there will also be inevitable hiccups associated with any new technology. Many manufacturers have already jumped on to the 3D bandwagon, and this includes big players like Panasonic, Sony, LG, Samsung and Mitsubishi. Also 3D is not a technology that will be exclusive to LCD panels. Plasmas and even the newer LED-

backlit displays will also get 3D love.

Initially we should see displays starting from 40-inches in size, and this should go all the way up to the larger 70-inch plasma panels.

How does it work?

All 3D displays work on the principle of stereoscopy. Remember that this is essentially a 3D image on a 2D display. This means to say that objects in a 3D scene (that is being viewed) will not change with respect to a viewer's perspective. In other words this is not like a 3D hologram, (except for intentional holographic displays), where a person with a difference perspective would see something different – this technology is still several years away from commercial use. Stereoscopy can very simply be defined as creating the illusion of 3D in a 2D image. The principle is called Stereopsis. It was first explained by the English scientist, Sir Charles Wheatstone as, "the mind perceives an object of three-dimensions by means of the two dissimilar pictures projected by it on the two retina". This is possible by tricking your mind and your eyes, by showing each eye a marginally different image. This different image per eye, tricks and forces your mind to merge these two images, thereby perceiving depth. This was illustrated by creating an illusion of a 3D image from 2D images that differed slightly in their point of view. The difference in the point-of-view is called horizontal disparity.

There are different technologies used to achieve a 3D effect. Broadly speaking these can be divided into two categories. The first category consists of those technologies where some sort of 3D glasses, are required to be worn. The other category consists of a developing technology that does without these glasses, and such displays are called auto-stereoscopic displays.

3D displays with glasses: Windows to the 3D world

3D glasses are of two types based on the type of lens used – active and passive lens. Passive lens are obviously cheaper, an example of a passive lens would be the simple glasses made of cardboard and feature two of different colours – one for each eye. Such glasses with lens